

Predicting Home Sale Prices using Linear Regression: Reflection & Lessons Learned



Presented by: Raven Otero-Symphony
CNM Ingenuity: Data Science Bootcamp
17 April 2026

Agenda

→ Problem Statement /section 1

→ Regression Assumptions /section 2

→ Exploratory Data Analysis /section 3

→ Model Results /section 4

→ (Near) Future Development /section 5

Problem Statement

- **4 years of housing data:** The fictional Accessor's Office has provided data on individual residential properties sold from 2006-2010
- **Data dictionary provided:** A document containing the names, data types, and values for each feature of the dataset is provided for context

Goal: Predict housing sales prices using linear regression, feature scaling, regularization, and minimizing the Root Mean Square Percentage Error (RMSPE) for reliable predictions



Linear Regression Assumptions

01

Linearity

Independent and dependent variables have a linear relationship.

02

Homoscedasticity of Residuals

To avoid unreliable coefficient estimates, there must be randomness in the variance of residuals.

03

Normal Distribution

Residuals must follow a normal distribution where multiple predictors are involved.

04

Independence of Errors

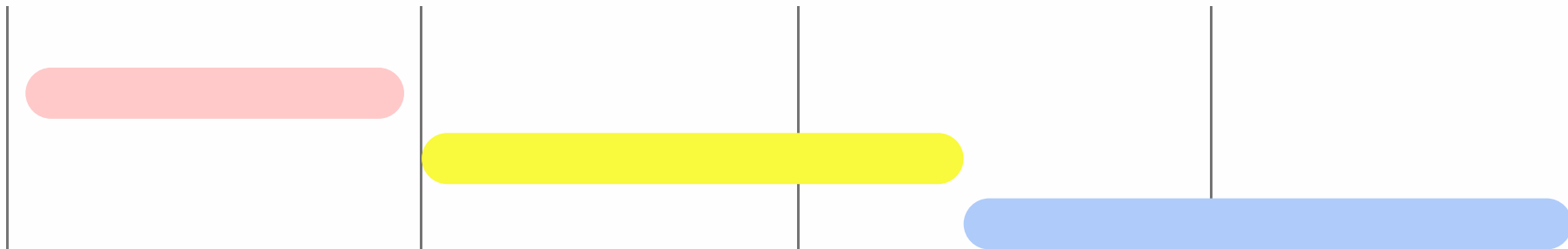
Residuals must not correlate with each other across observations.

05

Absence of Endogeneity

Residuals cannot be correlated with the error term.

The Plan:



01

Phase 1

Initial Data Analysis

- Familiarity with original data
- Read original data dictionary
- Check assumptions early
- Note significant observations

02

Phase 2

Exploratory Data Analysis

- Clean the data which could include:
 - Dropping null data
 - Imputing missing values
 - Transforming skewed features
 - One-hot and label encoding

03

Phase 3

Model Fitting & Results

- Fit an original regression model
- Repeat the process to reduce RMSPE, testing different hypotheses and communicating final results

What Happened



01

Phase 1 *Initial Data Analysis*

- I know what I'm doing, I got this
- Ok, not bad
- Hm, that's a little weird...
- Wait, this might actually be real data lol

02

Phase 2 *Exploratory Data Analysis*

- Who wrote this dictionary
- Should I write a whole new dictionary?
- Let's parse this into categories...
- Which section was I on?
- Oh shoot... I'm 80% through and didn't notice THAT...

03

Phase 3 *Model Fitting & Results*

- Fit the original model—wait, I can't because of categorical data
- Ok fine fit another one—wait, why didn't that code work?
- I have to restart the notebook... AGAIN...
- How is it 12am

Due to discrepancies in the data dictionary, the data was parsed and cleaned in 4 categories. 

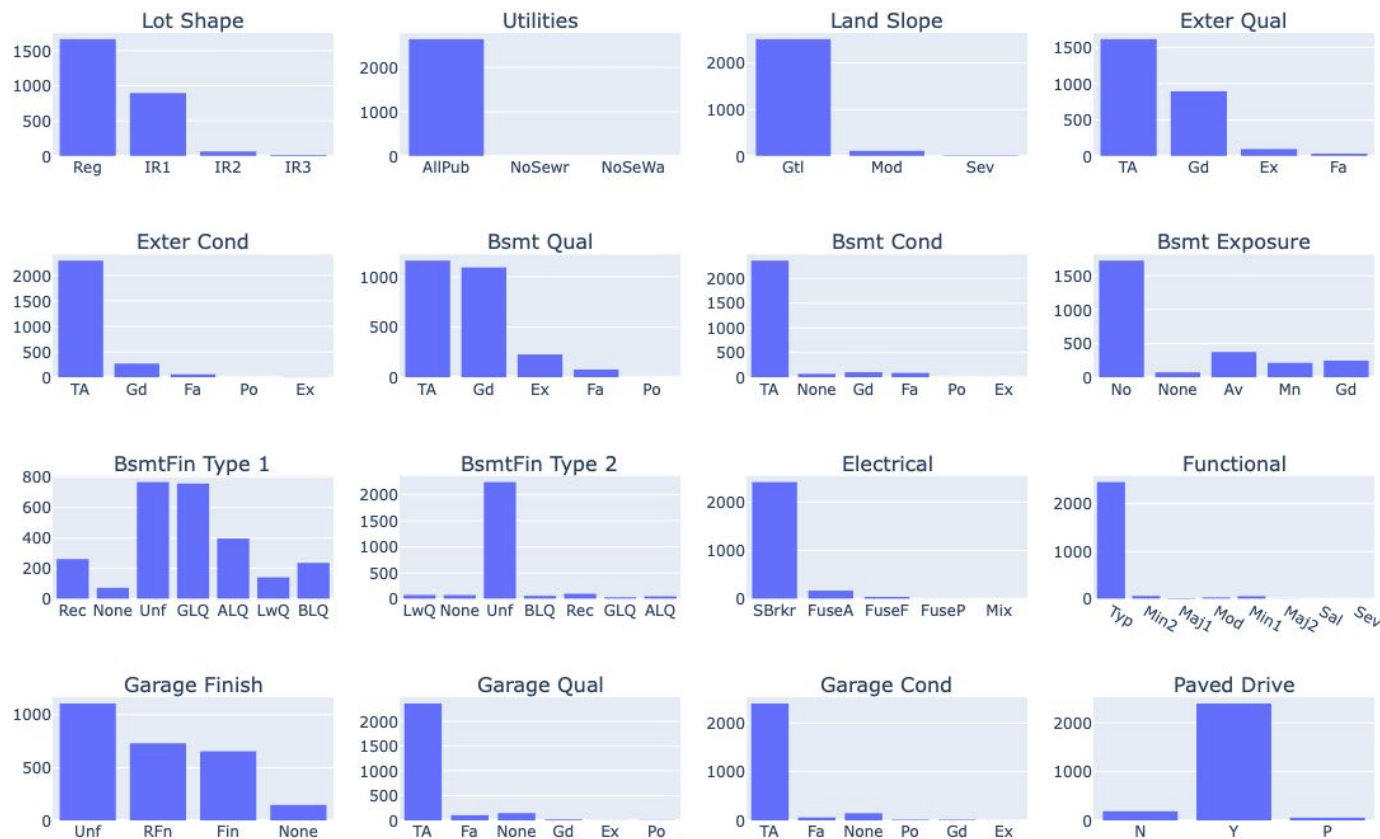
The original data contains **2,637 rows** and **81 features**.

However, the dictionary reports an additional **293 rows** and **1** feature, with data getting mistranslated in Python. 

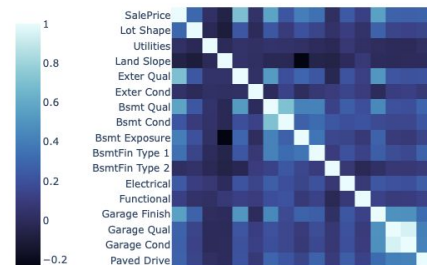
Ultimately, I chose to opt for the actual data that I was working with, ending up with **2,637 rows** and **111 features**.

Let's walk through it →

Ordinal Column Distributions



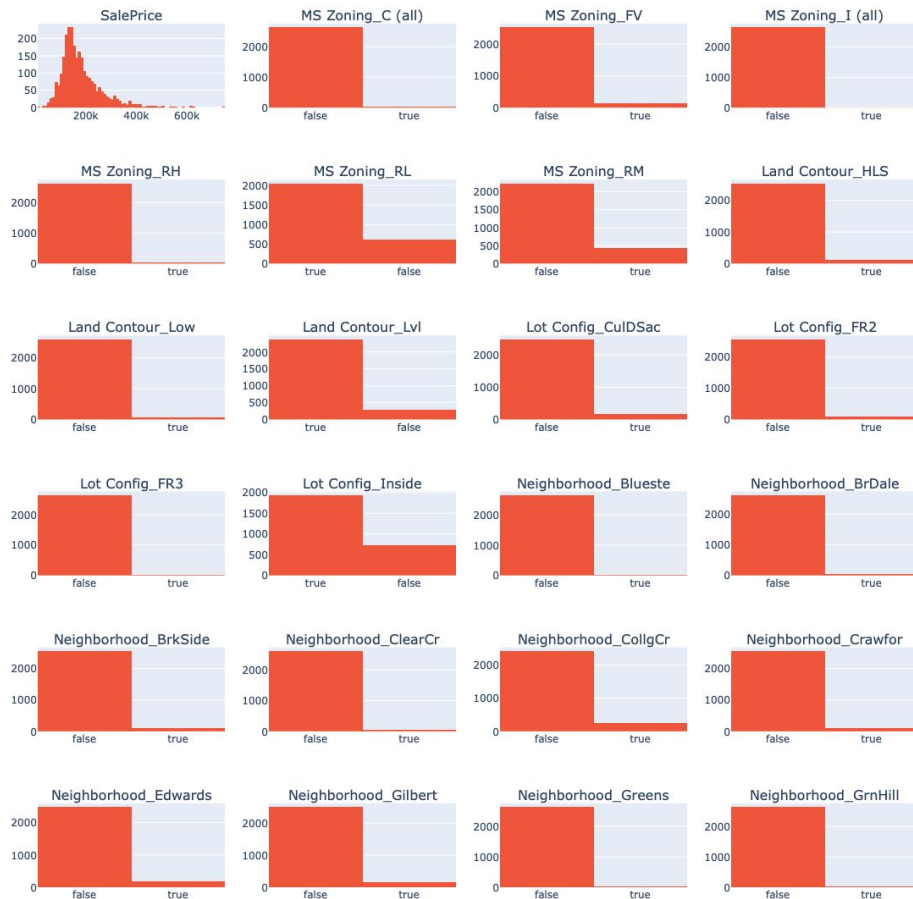
Ordinal Correlation Matrix with Target Variable



Ordinal Summary:

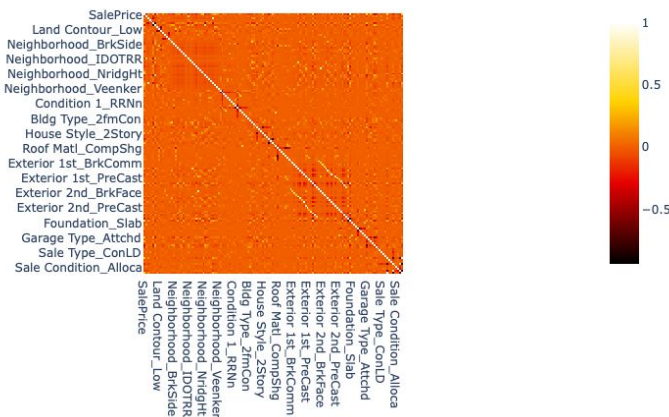
- 16 ordinal features
- Many are imbalanced
- Label encoded
- Defined an ordinal dictionary and applied `for` loop
- Kept **10 features** after correlation (cutoff 50%)

Nominal Column Distributions



Note: SalePrice is not nominal and was included for the correlation plot; disregarded in above

Nominal Correlation Matrix with Target Variable



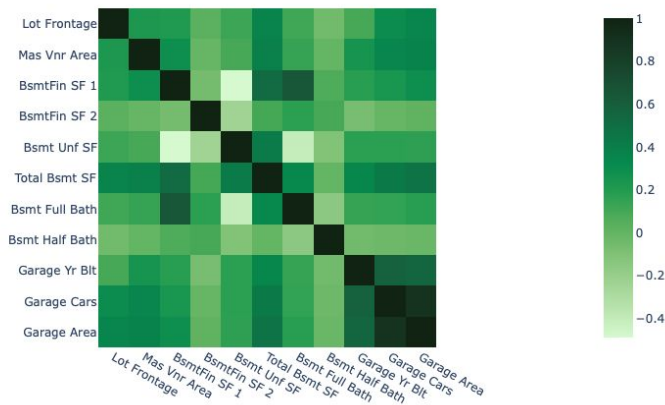
Nominal Summary:

- 21 original features
- Incorrectly named features (Exterior 1st & 2nd) from dictionary
- PID dropped as identifier
- Street and Central Air converted to boolean
- MS SubClass picked up as int, not categorical
- One-hot encoded
- Kept **87 features** after correlation (cutoff 50%)

Numeric (float64) Column Distributions



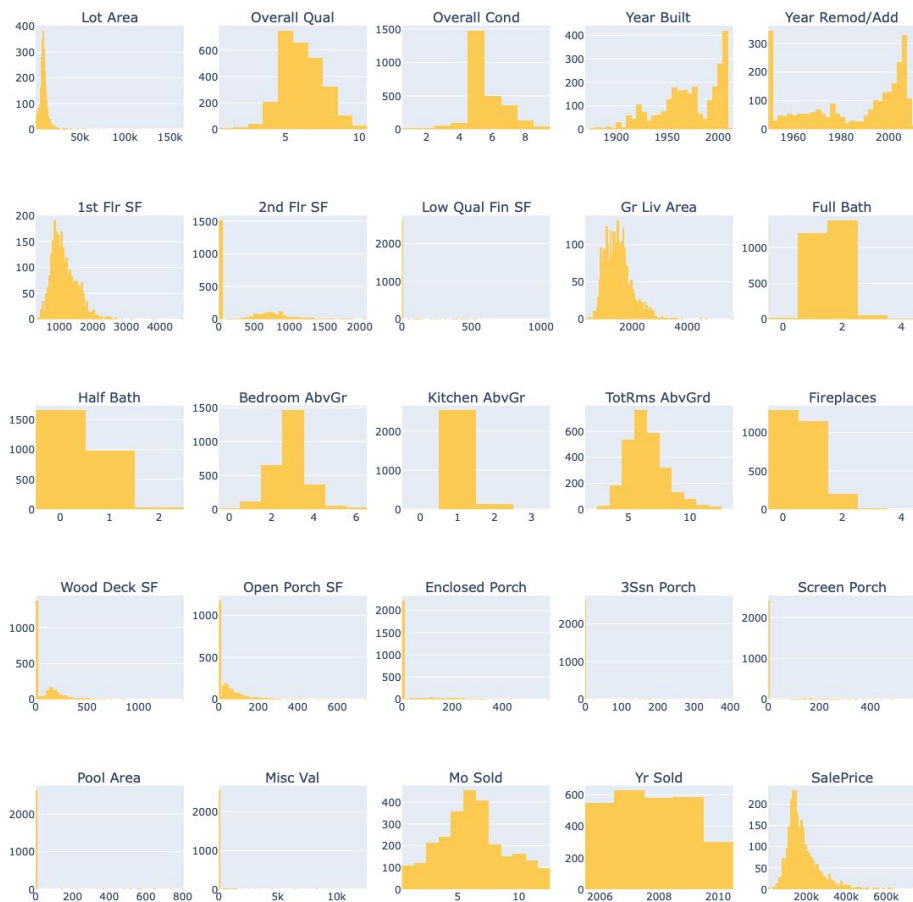
Numeric (float64) Correlation Matrix with Target Variable



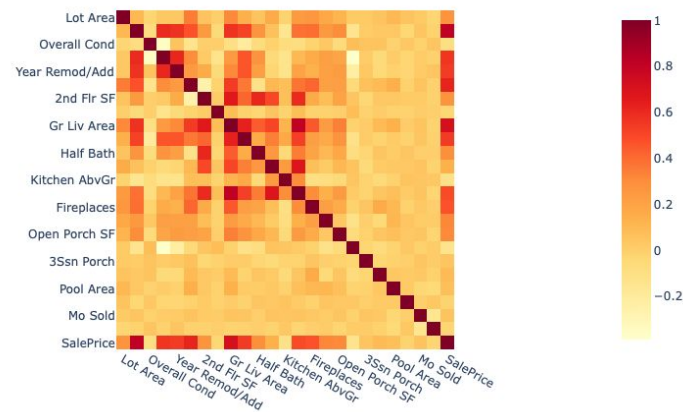
Numeric (float64) Summary:

- 11 original features
- **Garage Yr Blt** has a year value in 2207... I knew time travel was real...
 - This value needs to be dropped, but I ran out of time
- Highly right-skewed data, but ran out of time to transform
- Kept **5 features** after correlation (cutoff 50%)

Numeric (int) Column Distributions



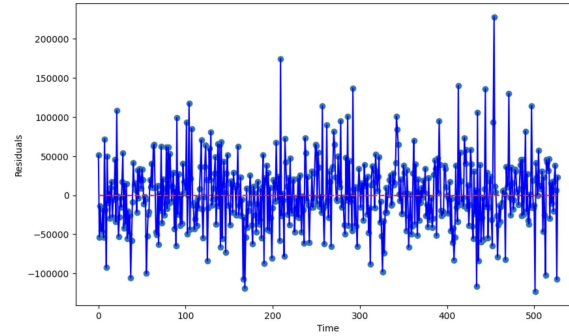
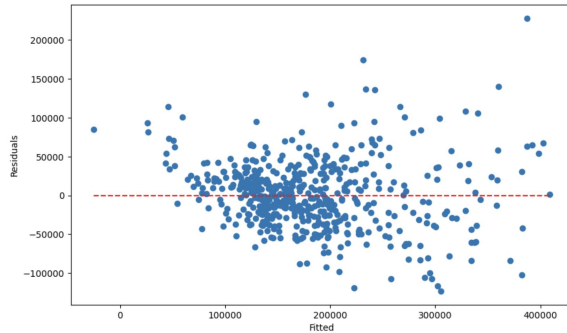
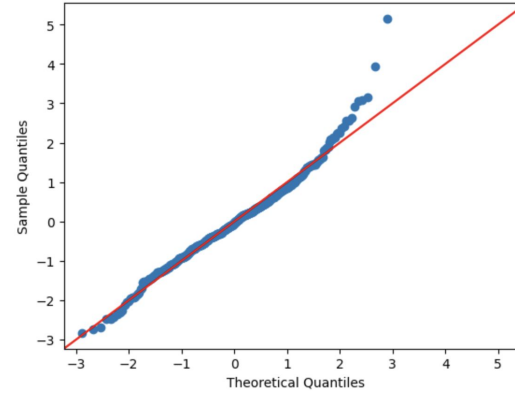
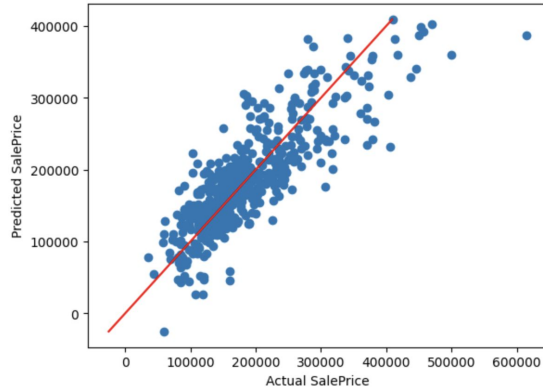
Numeric (int) Correlation Matrix with Target Variable



Numeric (int) Summary:

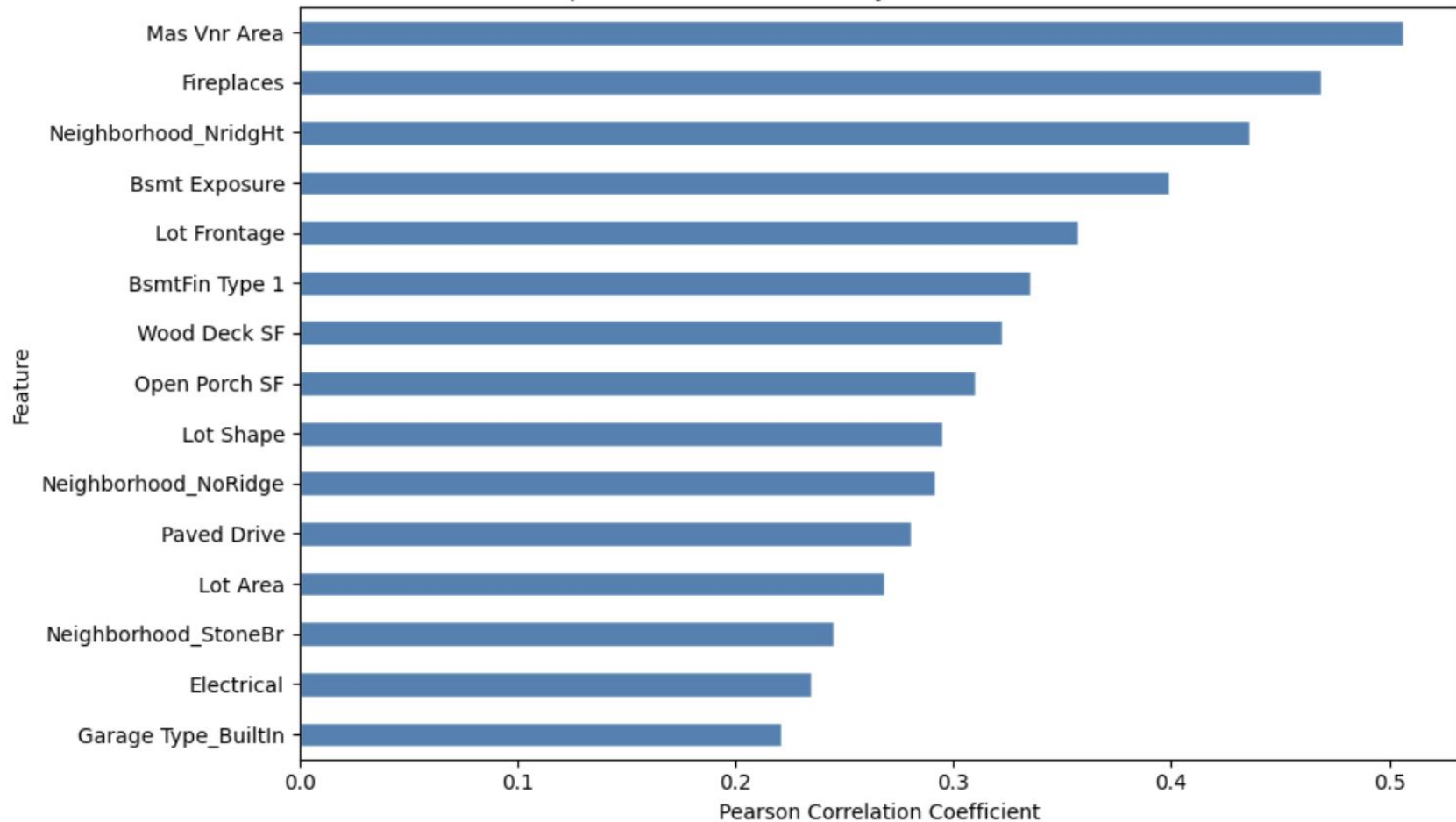
- 25 original features
- Highly right-skewed data, but ran out of time to transform
- **Year Built** and **Year Remod/Add** are dates, not ints
- Kept **14 features** after correlation (cutoff 50%)

Final Shape: 2637 x 111



RMSE: \$43,725 | RMSPE: 27%

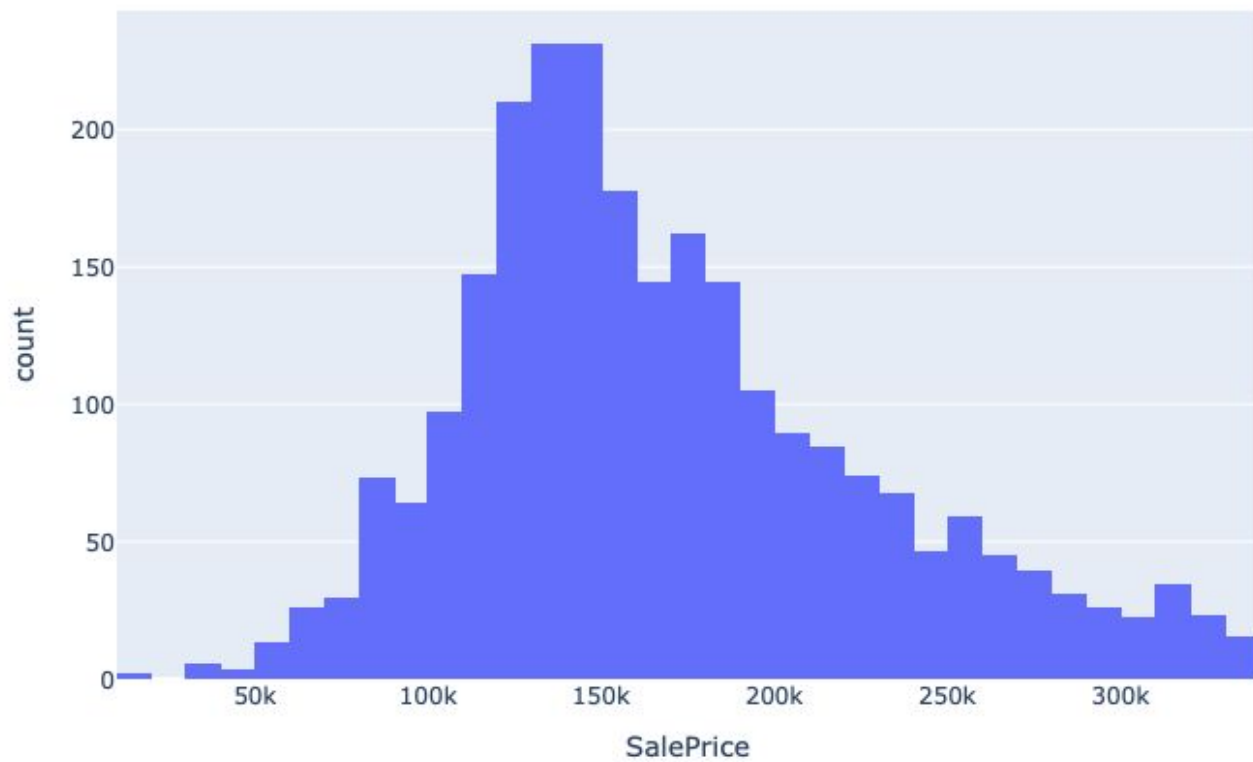
Top 15 Numeric Features by Correlation with SalePrice



(Near) Future Development

- Identify and remove outliers using IQR or similar method Issue #1
- Transform heavily-skewed features Issue #2
- Spend more time on model refinement Issue #3
- Adjust correlation cutoff and test other models to improve RMSPE Issue #4
- **New skills:** .set(), ordinal labeling, if/else to check column overlap, and for loops for print() and Plotly outputs Level Up! 🚀

IQR Outlier Removal for SalePrice



Thank you!



A new version will be posted soon

[[View Code](#)]